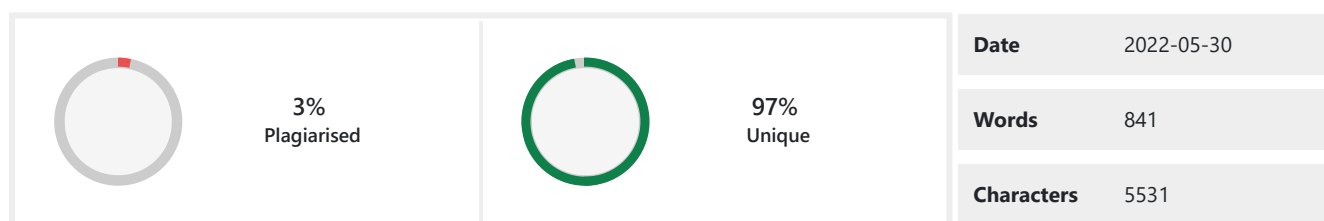


PLAGIARISM SCAN REPORT



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This model helps us to understand where things go haywire in our designs. Under the context of Industry 4.0, there are issues when it comes to the sections of machine execution and problem evaluation by the cyber-physical systems. Machine perceive the world in binaries, as opposed to sensory inputs that are so readily perceived by humans. There are also issues with the interpretation of perception which further causes issues with the evaluation of those interpretations. There is a measurable difference between user actions and those that a particular system can perform. A potent and engaging interface allows a user to perform an action without system limitations. This is especially important for Industry 4.0 which is nothing but an interconnected web of system of subsystems consisting of IOT and cyber-physical systems. The second consideration wherein the importance of having a proper interaction mechanism between humans and machine for cyber physical systems is the difference between the presentation of an output and the user's expectations. An effective interface can be easily evaluated by a user but it fails to evaluate when only machines are involved in the process. As a preliminary step, research questions were chalked out to define the basis of this research paper. The focus of these research questions is to primarily study the current visions of an Industry 4.0 ecosystem and especially explore this industrial revolution from the perspective of humans. Answering the formulated research questions would provide an overarching view of the human-centred approach of building the next industrial revolution. For answering the research questions, domain-specific keywords from each research question were used to explore the current research work through Google Scholar. Google Scholar was preferred as the source of all research content, as it gave an expansive search result and encompassed research work published from a wide range of publication organizations, thus covering a broad spectrum of resources. Research material from a diverse set of publication houses - IEEE, Research Gate, Elsevier, Science Direct, and MDPI - were studied and analysed. Providing better user support whenever there is a tangible difference expected output and the output generated by the user can be a vital factor in deciding the end goal of an efficient cyber-physical system. Thus, we pose the following important questions and try to answer them with our study on the principles of HCI under the context of Industry 4.0:

3.1 The User Interface and User Experience Design Paradigm:

What facets of the UI-UX components of the Human-Computer Interaction be envisioned from an Industry 4.0 perspective?

3.2 The Interaction Paradigm:

How should the interaction mechanism between Humans and Machines be shaped under the context of Industry 4.0?

3.3 The Manpower-Machinery Paradigm:

How will manpower be trained to interact with, supervise, and maintain complex machinery of Industry 4.0?

4 Research Solutions

Complex machinery of the fourth industrial revolution can be quite overwhelming to work with, and with massive chunks of information with respect to each machine, and the interconnection of machines, obtaining the right information at the

right time can be quite difficult. Information management and control over such information then become critical to avoid chaos. A carefully designed and well-placed HCI would help to steer clear of such a situation. The meticulous design and information display of an HCI system can be achieved by incorporating design process elements such as qualitative and quantitative analysis, and information architecture. For example, Gorecky et al. [25] have envisioned the HCI in Industry 4.0 to contain Augmented and Virtual Reality elements. They further stress that this inclusion of AR and VR, would come with a requirement for training to be given to employees working with Industry 4.0 machines, in which case the user interface should be intuitive and the user experience should be a good one. At the same time, employing technologies like AR/VR comes with its very own list of disadvantages. AR/VR headsets are difficult to wear for a long duration of time, and would cause an uncomfortable experience.

With heavy Head-Mounted Displays (HMDs) or limited battery capacities, hand-held or body-worn devices still challenge the worker, who is using it for eight hours a day [26].

We recommend for implementing the AR/VR correctly into the Industry 4.0 ecosystem, users need to be considered from the start in the design process of these interactions, to understand them and unearth their concerns and pain points through qualitative analysis such as interviews. These concerns should be further validated with a quantitative study such as surveys to realize a hypothesis. This will enable to gauge the value of introducing AR and VR at the initial level, and will also help in making modifications along the way, according to the best fit scenario with respect to industry personnel.

To answer the aforementioned questions on what recommendations or solutions can be conceived to have a seamless UI-UX component, interaction mechanism and how the manpower-machinery paradigm should work in the HCI which is specifically developed for and from an Industry 4.0 perspective, we define the following 6 components of the HCI centered on the Industry 4.0 realm:

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